

Clock enable vs gated clock

To save power or skip cycles you can stop updating a register, but how you stop matters

Load, hold, glitch

- Starter: clock-enable mode with `ce` equals one, each posedge loads `d` into `q`
- Set `ce` to zero and `q` holds even if `d` changes
- Switch to AND-gated mode and arm an enable glitch while `clk` is high, the warning lights up
- ICG sketch shows latched enable for cleaner gating
- Preferred RTL default: clock enable on the data path, not raw AND on `clk`

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Clock enable vs gated clock

INTERACTIVE TOOL

Clock enable vs gated clock

Prefer if (ce) q <= d on a free-running clock. AND-gating the clock risks glitches — know when ICG is the exception.

Starter example: clock-enable style — free-running clk, load with if (ce). Compare AND-gated clock (glitch risk) and ICG sketch.

Load starter example

Challenges (1/22)

Quiz: CE. Preferred style uses a? Answer: clock enable

Your answer

Show hintCheckNextIdle

quiz-cequiz-gatequiz-icgquiz-glitchstarterpreset-cepreset-gatedpreset-icgtoggle-cesteploadholdglitchdemoexplainmode-gatedcode-cecode-gatedcode-icgwarn-gatedtoggle-dfull-path

Core ideas

Clock enable

Mux/hold on D — clock stays clean and free-running.

Gated clock

AND on clk — glitches & skew if enable is raw combo.

ICG

Library clock-gate cell when you truly must gate.

Clock enable starter

learn_digital — clock enable

Workbook practice

- In the workbook track, write the always block for clock enable versus AND-gated clock
- Sketch clk, ce, and q for CE equals one then CE equals zero with d changing
- Explain why ce toggling during clk high is risky for gated mode
- Name one pitfall: using AND-gated clocks without an ICG cell in real silicon

Pitfalls to watch

- Do not confuse holding q with stopping the clock, CE holds data; gating stops edges
- Glitches on gated clocks can cause extra captures or timing nightmares
- And remember: the browser lab is literacy
- Real designs still need clock-tree constraints, ICG rules, and STA on enable paths

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one CE hold wave and one gated-glitch caution
- When you are ready, take the short quiz, then continue to CDC and two-FF sync